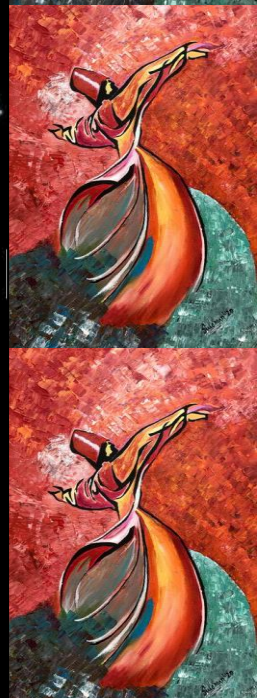




EE 354/CE 361/MATH 310 -
Introduction to Probability
and Statistics
Spring 2026



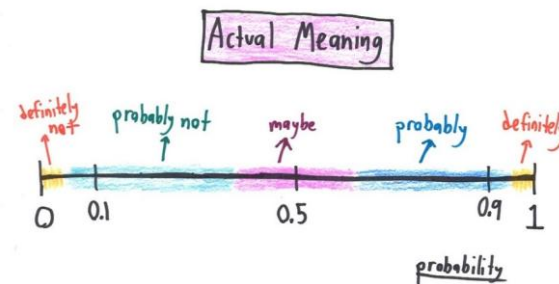


EE 354/CE 361/MATH 310 L4 section (Spring 2026)

Introduction to Probability and Statistics -

Aamir Hasan

Week 13 - Lecture No 24 – April 06, 2026



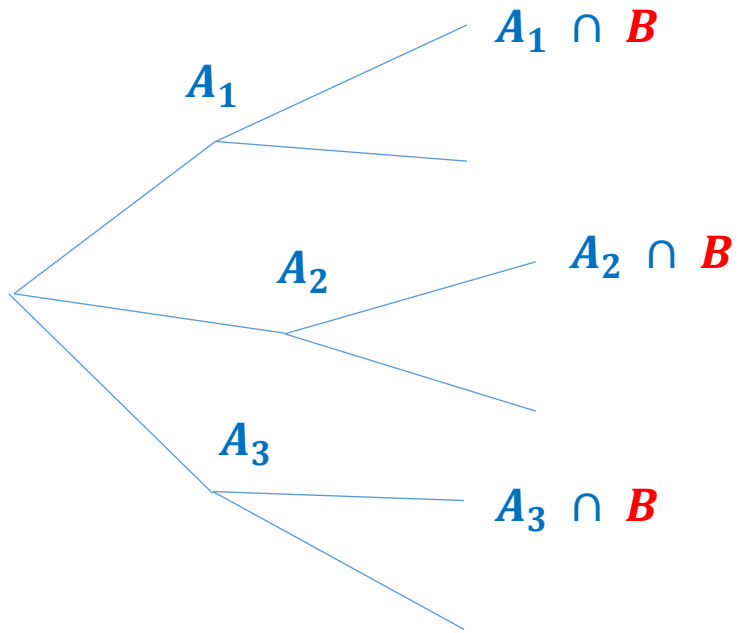
Announcements!

1. Office hours (post Ramadan) – Tuesday & Thursday 12:00 to 1:00 PM
2. Planned quizzes
 1. Quiz No 07 – April 08, 2026 (Wednesday)
 2. Quiz No 08 – April 15, 2026 (Wednesday)
 3. Quiz No 09 – April 22, 2026 (Wednesday)
 4. Quiz No 10 – April 29, 2026 (Wednesday)

Agenda for today

1. Unit 5 – Continuous Random Variable
 - Recap – Total Probability Theorem & Total Expectation Theorem.
 - Mixed Distribution

Total probability and expectation theorems



$$P(B) = P(A_1) P(B | A_1) + \dots + P(A_n) P(B | A_n)$$

$$p_X(x) = P(A_1) p_{X|A_1}(x) + \dots + P(A_n) p_{X|A_n}(x)$$

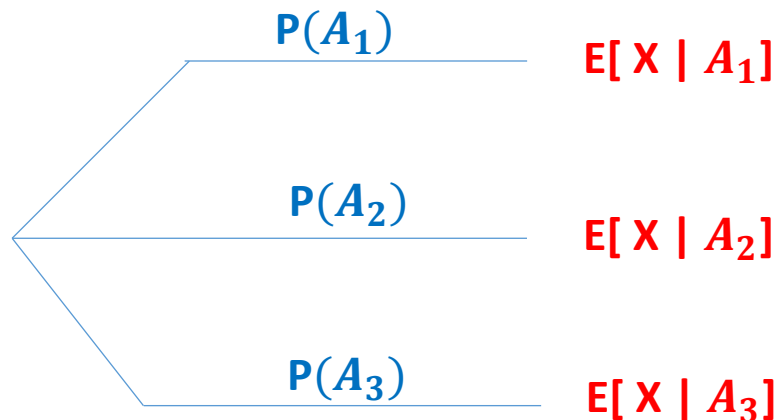
$$F_X(x) = P(X \leq x) = P(A_1) P(X \leq x | A_1) + \dots$$

$$= P(A_1) F_{X|A_1}(x) + \dots$$

$$f_X(x) = P(A_1) f_{X|A_1}(x) + \dots + P(A_n) f_{X|A_n}(x)$$

$$\int x \cdot f_X(x) dx = P(A_1) \int x \cdot f_{X|A_1}(x) dx + \dots$$

$$E[X] = P(A_1) E[X|A_1] + \dots + P(A_n) E[X|A_n]$$



Example

Sameena goes to the supermarket shortly, with probability $1/3$, at a time uniformly distributed between 0 and 2 hours from now; or with probability $2/3$, later in the day, at a time uniformly distributed between 6 to 8 hours from now. Find the pdf $f_X(x)$, where RV X is the time of departure to the supermarket. Also, calculate $E[X]$.

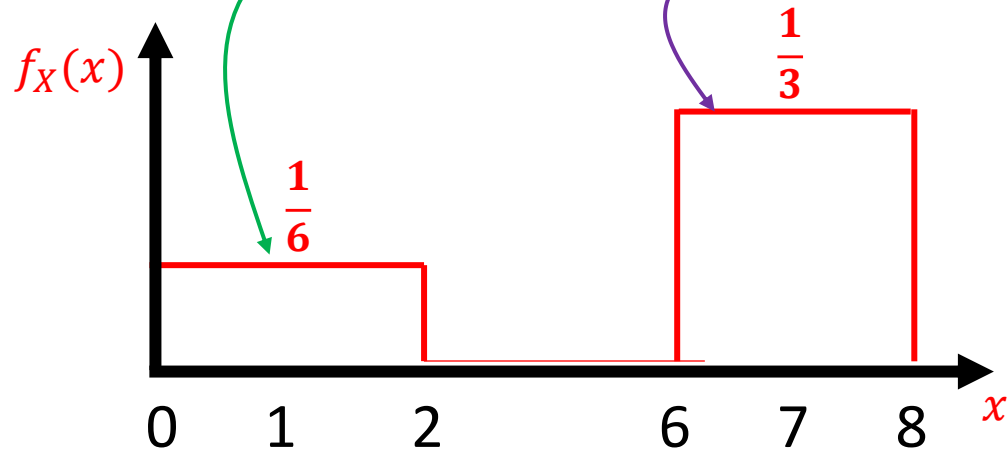
$$f_{X|A_1} \sim U[0, 2]$$

$$f_{X|A_2} \sim U[6, 8]$$

$$P(A_1) = \frac{1}{3}$$

$$P(A_2) = \frac{2}{3}$$

$$f_X(x) = P(A_1)f_{X|A_1}(x) + \dots + P(A_n)f_{X|A_n}(x)$$



$$E[X] = P(A_1) \cdot E[X|A_1] + \dots + P(A_n) \cdot E[X|A_n]$$

$$\frac{1}{3} * 1 + \frac{2}{3} * 7$$

Mixed Distributions

You play lottery and you have 1 dollar. You win dollars X : -

$$X = \begin{cases} 1, & \text{with probability } \frac{1}{2} \\ \text{uniform on } [0, 2], & \text{with probability } \frac{1}{2} \end{cases}$$

Is X Discrete? **No**

Is X Continuous? **No**

Y Discrete
 Z Continuous

$$X = \begin{cases} Y, & \text{with probability } (1 - p) \\ Z, & \text{with probability } p \end{cases}$$

$$P(X=1) = \frac{1}{2}$$

X is mixed

$$\rightarrow F_X(x) = P(X \leq x)$$

$$\rightarrow F_X(x) = (1-p)P(Y \leq x) + pP(Z \leq x)$$

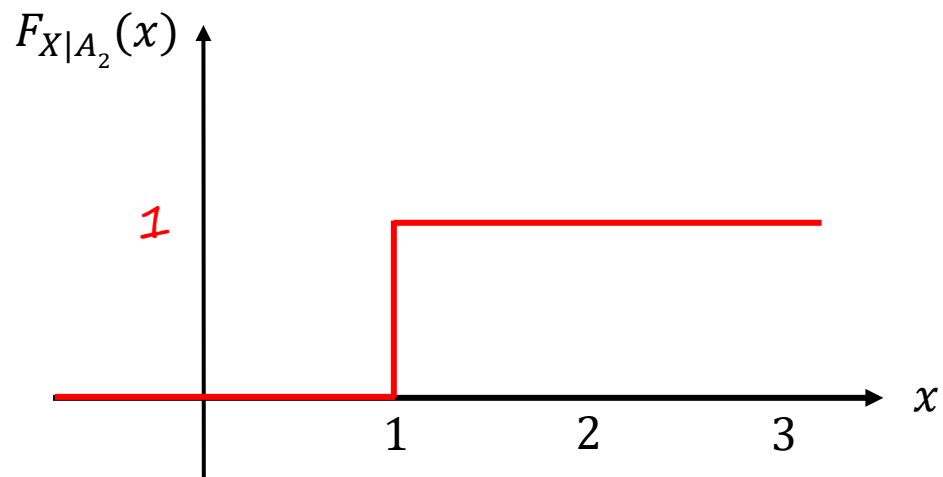
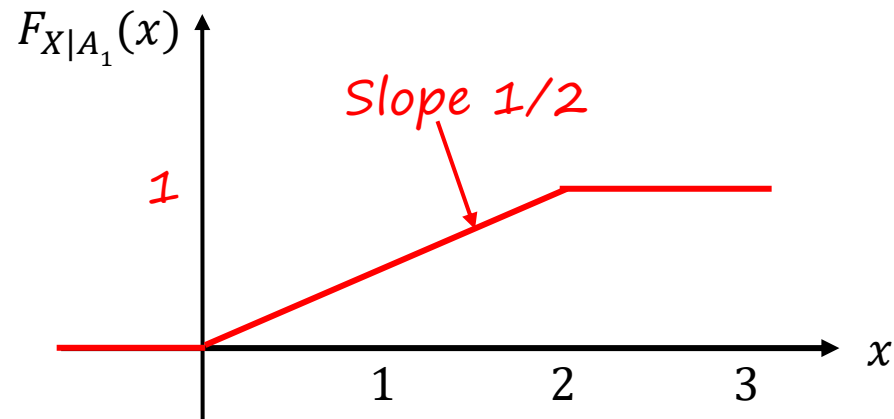
$$\rightarrow = (1-p)F_Y(x) + pF_Z(x)$$

$$\rightarrow E[X] = (1-p)E[Y] + pE[Z]$$

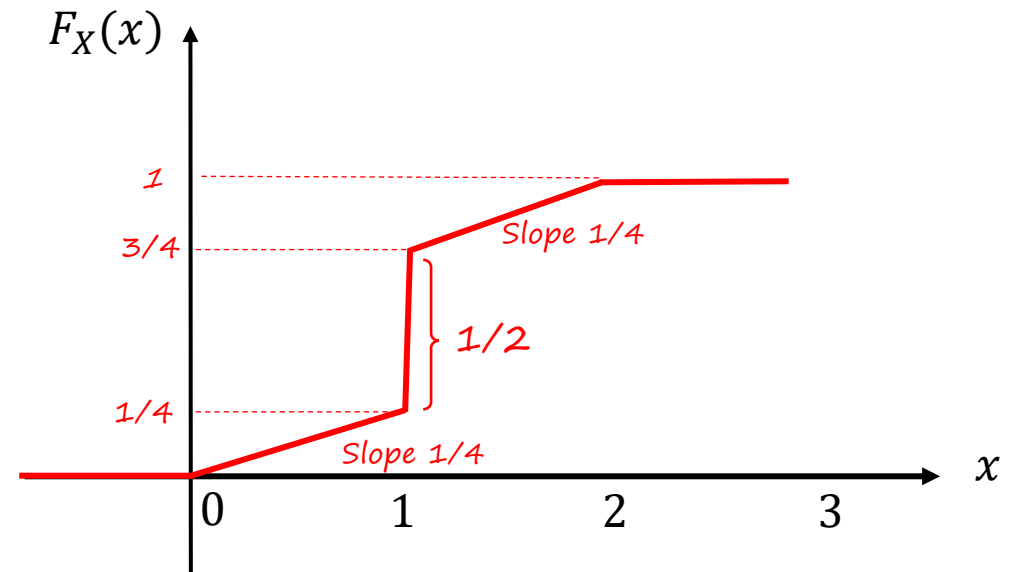
Mixed Distributions

$$X = \begin{cases} 1, & \text{with probability } 1/2 \text{ } \mathbf{A2} \\ \text{uniform on } [0, 2], & \text{with probability } 1/2 \text{ } \mathbf{A1} \end{cases}$$

**Determine &
plot $F_X(x)$**



$$F_X(x) = \mathbf{P}(A_1) F_{X|A_1}(x) + \mathbf{P}(A_2) F_{X|A_2}(x)$$



Practice Problem - Mixed Distributions

Question 1 [20 points] Cumulative Distribution Function for Mixed Distribution

- **Discrete Random Variable X** - You toss a biased coin two times, where the **RV X** is the number Heads in the two throws. Therefore, X is a Binomial Random Variable with parameter $n = 2$ with probability of success (Heads) = $1/3$.
 - **Continuous Random Variable Y** – RV Y has a uniform distribution with $0 \leq y \leq 3$.
 - **Random Variable Z** = X with probability 0.5 and Z = Y with probability 0.5. Random Variable Z has a mixed distribution.
- (a) **[8 points]** Calculate and plot the Cumulative Distribution Function (CDF) for RV Z. *Use the diagram given below (Figure 1) to plot the CDF, clearly marking the exact CDF heights.*
- (b) **[4 points]** Find the probability that RV Z takes on values in the interval, $1 \leq z \leq 2$.
- (c) **[6 points]** Let's now assume that the Continuous Random Variable Y is a standard Gaussian RV with 0 mean and unit variance, i.e. $Y \sim N(0, 1)$. Again, Random Variable Z = X with probability 0.5 and Z = Y with probability 0.5. Calculate and plot the Cumulative Distribution Function (CDF) for RV Z. *Use the diagram given below (Figure 2, Page 2) to plot the CDF, clearly marking the CDF heights (be precise as much as you can).*
- (d) **[2 points]** Find the probability (for the revised RV Y as in Part c) that the RV Z takes on values in the interval, $-\infty \leq z < 0$.

Solution

Practice Problem – Mixed Distributions

$$P_X(0) = \left(\frac{2}{3}\right)^2 = 4/9, \quad P_X(1) = 2 \cdot \frac{1}{3} \cdot \frac{2}{3} = 4/9$$

$$P_X(2) = 1/9$$

$$F_X(0) = 4/9, \quad F_X(1) = 8/9, \quad F_X(2) = 1$$

$$F_Z(z) = \frac{1}{2} \cdot F_X(z) + \frac{1}{2} F_Y(z)$$

$$F_Z(0) = \frac{1}{2} \cdot 4/9 = 2/9 = 0.222$$

$$F_Z(1) = \frac{1}{2} \cdot 8/9 + \frac{1}{2} \cdot 1/3 = 0.61$$

$$F_Z(2) = \frac{1}{2} + \frac{1}{2} \cdot 2/3 = 0.833$$

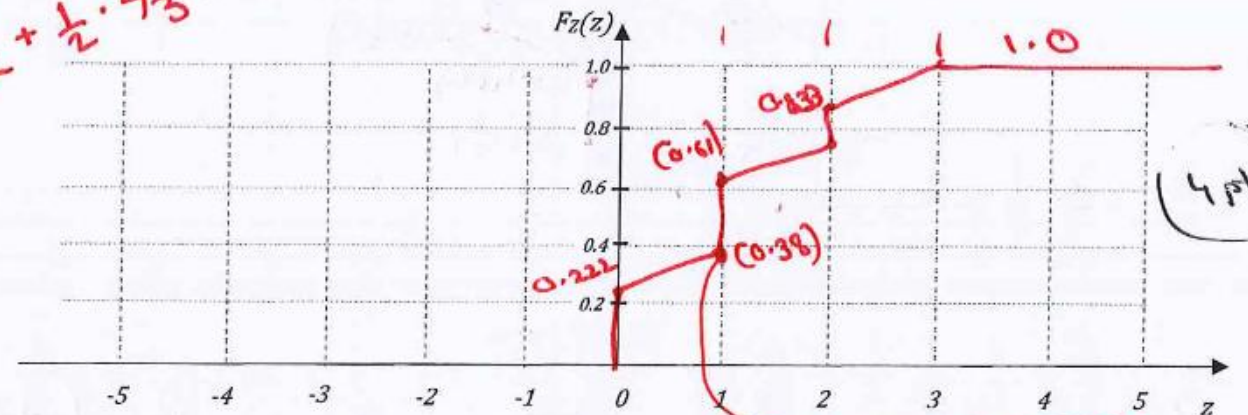
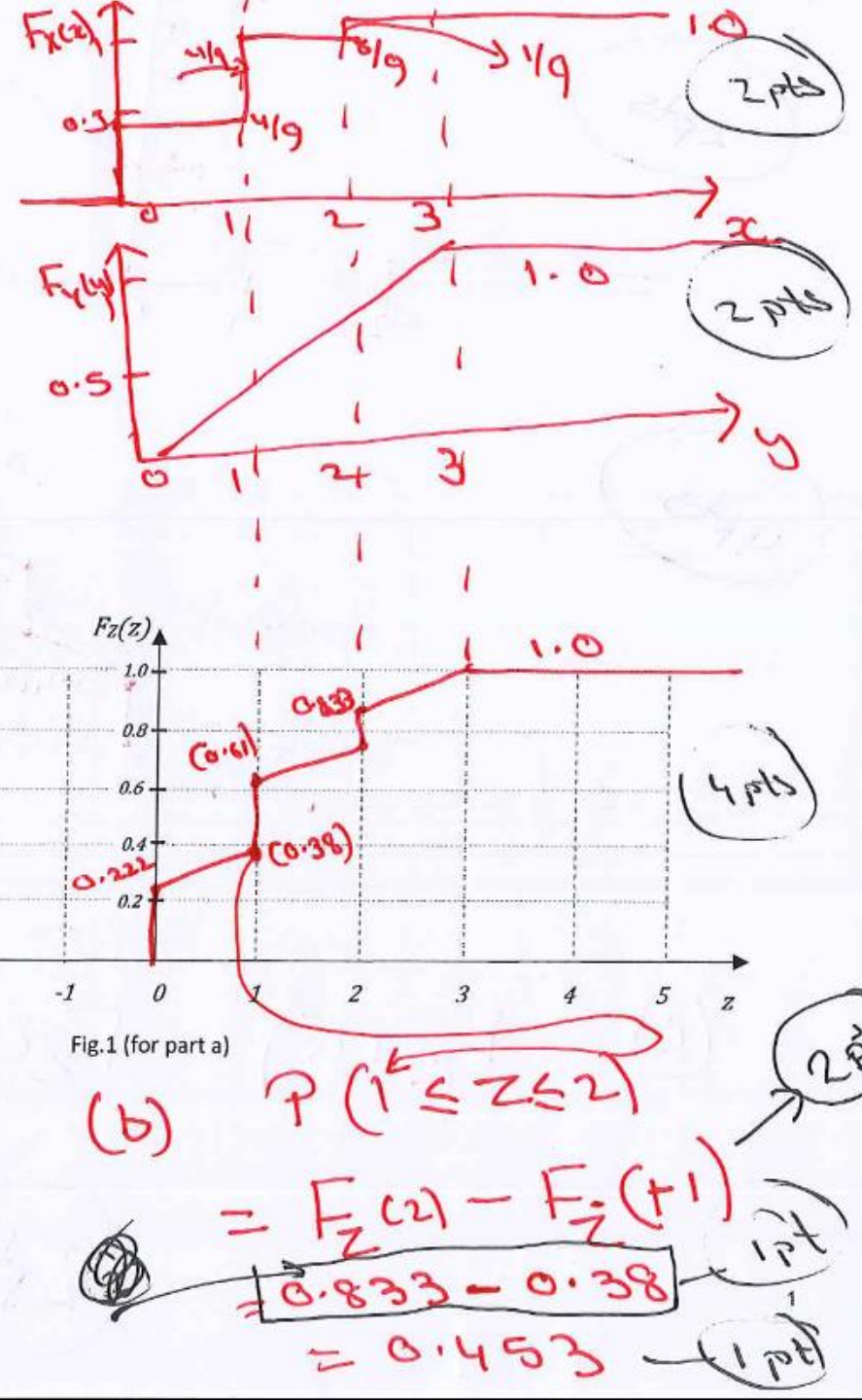


Fig.1 (for part a)

$$F_Z(3) = \frac{1}{2} \cdot 1 + \frac{1}{2} \cdot 1 = 1$$

(b) $P(1 \leq Z \leq 2)$

$$= F_Z(2) - F_Z(1)$$

$$= 0.833 - 0.61 = 0.223$$

The final result 0.223 is boxed and labeled with '1 pt' and '2 pt' in circles.

Solution

Practice Problem – Mixed Distributions

(c) $F_X(0) = 4/9$, $F_X(1) = 8/9$ & $F_X(2) = 1.0$

$F_Y(-2) = 0.022$, $F_Y(-1) = 0.1587$, $F_Y(0) = 0.5$, $F_Y(1) = 0.8413$
 $F_Y(2) = 0.9772$

$F_Z(z) = \frac{1}{2} \cdot F_X(x) + \frac{1}{2} F_Y(y)$ $F_Y(3) \approx 1$

$F_Z(-2) = 0.0114$, $F_Z(-1) = 0.07935$, $F_Z(0) = 0.4722$
 $F_Z(1) = 0.865$, $F_Z(2) = 0.9886$, $F_Z(3) = 1$

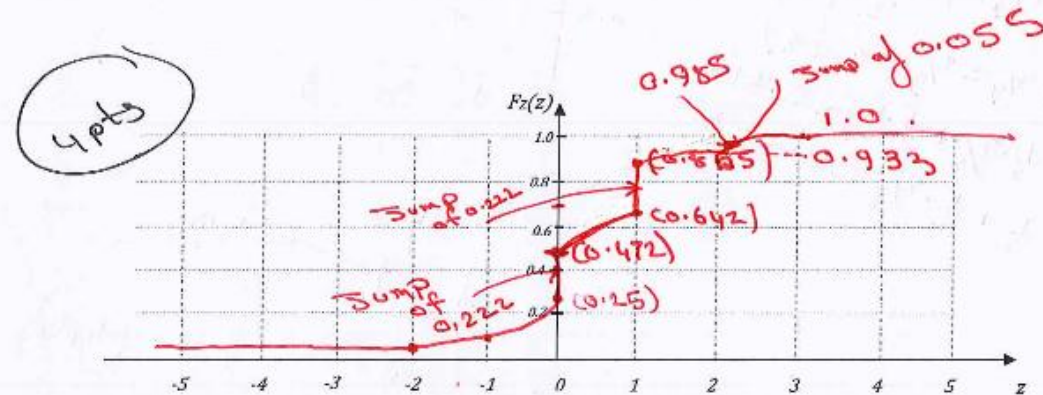
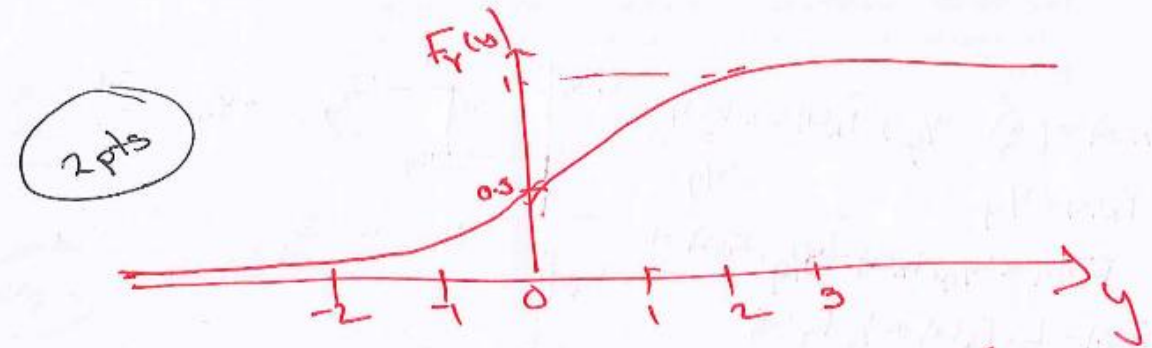


Fig. 2 (for part c)

(d) $F_Z(0^-) - F_Z(-\infty) = P(-\infty \leq Z < 0)$
 $= 0.25$

2 pts

Jointly continuous R.V.'s and Joint PDFs

r.v. X & Y

Jointly continuous R.V.'s and Joint PDFs.

$$p_X(x) \quad f_X(x)$$

$$p_{X,Y}(x, y) \quad f_{X,Y}(x, y)$$

$$p_{X,Y}(x, y) = \mathbf{P} (X = x \text{ and } Y = y) \geq 0$$

$$f_{X,Y}(x, y) \geq 0$$

$$\mathbf{P}((X, Y) \in B) = \sum_{(X, Y) \in B} \sum p_{X,Y}(x, y)$$

$$\mathbf{P}((X, Y) \in B) = \iint_{(X, Y) \in B} f_{X,Y}(x, y) \, dx dy$$

$$\sum_x \sum_y p_{X,Y}(x, y) = 1$$

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) \, dx dy = 1$$

Definition:

Two random variables are **jointly continuous** if they can be described by a joint PDF